

PAPER_1366 — Bird Magnetoreception via Cryptochrome

Framework: UQFF — Star-Magic v5.27+ | **Tier:** G Bio/Complex (CLOSED) **Calculator:** `calculate_paradox({"paradox": "magnetoreception"})`

Master Expression

$\text{threshold} = \beta_i \times \Phi_{\text{res}} = 50.6\% \text{ Earth field}$

Match: sensitivity match

Interpretation

Radical pair mechanism in cryptochrome via SCm coupling. Detection threshold ~50% of Earth field ($50 \mu\text{T} \times 0.506$).

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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